TOPIC 3 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

Second-Order Linear Equations: General Theory, Vibrations and Forced Response

The theory that governs every linear vibrating system: why two initial conditions are needed, why two independent solutions are enough, how the Wronskian decides independence, how to solve constant-coefficient equations in all three root cases, and how to find particular solutions by undetermined coefficients, reduction of order and Lagrange's variation of parameters.

Ground truth: Robinson (2004) Ch. 11 (pp. 101–110), Ch. 12 (pp. 111–119), Ch. 13 (pp. 120–130), Ch. 14 (pp. 131–140), Ch. 15 (pp. 141–149), Sec. 16.1 (p. 150), Ch. 17 (pp. 159–163), Ch. 18 (pp. 164–168) • Lecture map, Parts II–III (Sep 22 – Oct 1)

≈ 30 min study time • 36 displayed equations • 4 figures

SECTION 01 • ROBINSON SEC. 11.1–11.2 • LECTURE SEP 22

1. Second-order equations: two initial conditions

Newton's second law is second order: the force fixes the acceleration $\ddot{x}$, not the velocity. A general second-order equation reads [Robinson, Eq. (11.1), p. 101]

$$\frac{d^2x}{dt^2} = f(\dot{x}, x, t). \tag{11.1}$$

1.1 Why both $x(t_0)$ and $\dot{x}(t_0)$ are needed

Take the simplest case $\ddot{x} = f(t)$ (11.2), the motion of a particle under a prescribed force per unit mass. Integrate twice from t_0:

1. $\dot{x}(t) - \dot{x}(t_0) = \int_{t_0}^{t} f(s)\,ds$ (Barrow's rule on $\ddot{x} = \frac{d}{dt}\dot{x}$). One constant, $\dot{x}(t_0)$, appears.
2. $x(t) - x(t_0) = \int_{t_0}^{t} \dot{x}(s)\,ds = \dot{x}(t_0)(t - t_0) + \int_{t_0}^{t}\int_{t_0}^{s} f(r)\,dr\,ds$. A second constant, $x(t_0)$, appears.

Two integrations produce two arbitrary constants, so the position alone does not determine the motion: a piece of chalk can be dropped or thrown from the same point [p. 102].

THEOREM 11.1 — EXISTENCE AND UNIQUENESS [ROBINSON, P. 102]
Suppose $f(x_2, x_1, t)$, $\partial f/\partial x_1$ and $\partial f/\partial x_2$ are continuous for $a_1 < x_1 < a_2, b_1 < x_2 < b_2, t_1 < t < t_2$. Then for all initial conditions

$$x(t_0) = x_0, \qquad \dot{x}(t_0) = y_0 \tag{11.4}$$

with $a_1 < x_0 < a_2, b_1 < y_0 < b_2, t_1 < t_0 < t_2$, the equation $\ddot{x} = f(\dot{x}, x, t)$ has a unique solution, with two continuous derivatives, on some interval I containing t_0.

Here x_1 stands for x and x_2 for $\dot{x}$: the theorem is Theorem 6.2 applied to the equivalent first-order system $\dot{x}_1 = x_2, \dot{x}_2 = f(x_2, x_1, t)$ (Topic 4, Theorem 25.3). For *linear* equations with continuous coefficients the solution in fact exists on the whole interval where the coefficients are continuous (a standard result, not stated in the book); with constant coefficients, for all $t \in \mathbb{R}$ (Theorem 16.1, p. 150) – no blow-up is possible.

1.2 Linear equations and the operator L

A second-order linear equation $a_2(t)\ddot{x} + a_1(t)\dot{x} + a_0(t)x = g(t)$ with $a_2(t) \neq 0$ can be divided by a_2 to give the standard form [Eqs. (11.6)-(11.7), pp. 102-103]

$$\ddot{x} + p(t)\,\dot{x} + q(t)\,x = f(t) \quad \text{(inhomogeneous)}, \qquad \ddot{x} + p(t)\,\dot{x} + q(t)\,x = 0 \quad \text{(homogeneous)}. \tag{11.6–11.7}$$

Writing $L[x] = \ddot{x} + p\dot{x} + qx$ (11.10), the equations are $L[x] = f$ and $L[x] = 0$, and L is linear (11.11):

$$L[\alpha x_1 + \beta x_2] = \alpha L[x_1] + \beta L[x_2].$$

PRINCIPLE OF SUPERPOSITION [ROBINSON, EQ. (11.8), P. 103]
If x_1 and x_2 solve the homogeneous equation (11.7), so does $x = \alpha x_1 + \beta x_2$ for any constants α, β.

PROOF
Differentiation is linear: $\dot{x} = \alpha\dot{x}_1 + \beta\dot{x}_2$ and $\ddot{x} = \alpha\ddot{x}_1 + \beta\ddot{x}_2$. Therefore

$$\ddot{x} + p\dot{x} + qx = \alpha\big(\ddot{x}_1 + p\dot{x}_1 + qx_1\big) + \beta\big(\ddot{x}_2 + p\dot{x}_2 + qx_2\big) = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

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2. Linear independence and the general solution

DEFINITION 11.2 — LINEAR INDEPENDENCE OF FUNCTIONS [ROBINSON, P. 104]
The functions $x_1(t), \dots, x_n(t)$ are **linearly independent** on an interval I if the only solution of $\alpha_1 x_1(t) + \cdots + \alpha_n x_n(t) = 0$ for all $t \in I$ is $\alpha_1 = \cdots = \alpha_n = 0$. Otherwise they are linearly dependent.

For two functions this means "not proportional": if $x_2 = c x_1$ then $c x_1 - x_2 = 0$ is a non-trivial combination; conversely, if $\alpha_1 x_1 + \alpha_2 x_2 = 0$ with $\alpha_1 \neq 0$, then $x_1 = -(\alpha_2/\alpha_1) x_2$ [pp. 104–105]. So e^{t} and e^{2t} are independent, while $\sin 2t$ and $\sin t \cos t$ are not ($\sin 2t = 2 \sin t \cos t$).

THEOREM — TWO INDEPENDENT SOLUTIONS ARE NECESSARY AND SUFFICIENT [ROBINSON, SEC. 11.3.2, PP. 105–106]
The homogeneous equation $\ddot{x} + p\dot{x} + qx = 0$ has two linearly independent solutions x_1, x_2, and every solution can be written as $x(t) = \alpha x_1(t) + \beta x_2(t)$ for suitable constants α, β (11.12). One solution is not enough.

PROOF

Two independent solutions exist. By Theorem 11.1 there are solutions with $x_1(t_0) = 1, \dot{x}_1(t_0) = 0$ and $x_2(t_0) = 0, \dot{x}_2(t_0) = 1$. If $x_2 = c x_1$ then $x_2(t_0) = c$ forces $c = 0$, but then $\dot{x}_2(t_0) = 0 \neq 1$. So they are not proportional.

Two are enough. Let x_1, x_2 be any two independent solutions and x any solution, with $x(t_0) = x_0$, $\dot{x}(t_0) = v_0$. We want α, β with $\alpha x_1 + \beta x_2$ matching these data:

$$\begin{pmatrix} x_1(t_0) & x_2(t_0) \\ \dot{x}_1(t_0) & \dot{x}_2(t_0) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} x_0 \\ v_0 \end{pmatrix}. \tag{11.15}$$

This is solvable for every right-hand side if and only if the determinant $x_1(t_0)\dot{x}_2(t_0) - x_2(t_0)\dot{x}_1(t_0)$ is non-zero. Suppose instead it were zero (11.16). Then the columns are proportional: there is a constant c with $x_1(t_0) = c x_2(t_0)$ and $\dot{x}_1(t_0) = c\dot{x}_2(t_0)$ (11.17). (Robinson divides by $x_2(t_0)$ and $\dot{x}_2(t_0)$ to find c; they cannot both vanish, because then uniqueness would force $x_2 \equiv 0$, and if one vanishes the same conclusion follows directly – footnote, p. 106.) By superposition $y = c x_2$ is a solution, and it has the same data at t_0 as x_1. By uniqueness $x_1 = c x_2$ for all t, contradicting independence. Hence the determinant is non-zero, (11.15) has a solution, and by uniqueness $x = \alpha x_1 + \beta x_2$ everywhere.

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LINEAR-ALGEBRA READING (ROBINSON SEC. 11.5, STARRED)
L maps the vector space $C^2(I)$ of twice continuously differentiable functions to $C^0(I)$. The solution set of $L[x] = 0$ is its kernel, a vector space, and the theorem says $\dim \ker L = 2$ with basis $\{x_1, x_2\}$. For an n th-order equation, $\dim \ker L = n$ (Sec. 16.2).

2.1 The Wronskian [Robinson, Sec. 11.4, starred, pp. 106-107]

DEFINITION — WRONSKIAN

$$W[x_1, x_2](t) = \begin{vmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{vmatrix} = x_1(t)\,\dot{x}_2(t) - x_2(t)\,\dot{x}_1(t).$$

THEOREM — WRONSKIAN TEST (LECTURE, SEP 22; ROBINSON P. 107)

1. For any two differentiable functions: if $W[x_1, x_2](t_0) \neq 0$ at some $t_0 \in I$, then x_1, x_2 are linearly independent on I.
2. For two **solutions** of $\ddot{x} + p\dot{x} + qx = 0$: they are linearly independent if and only if $W \neq 0$; and then $W(t) \neq 0$ for every $t \in I$.

PROOF

(1) Suppose $\alpha x_1(t) + \beta x_2(t) = 0$ for all $t \in I$. Differentiating, $\alpha \dot{x}_1(t) + \beta \dot{x}_2(t) = 0$ too. At $t = t_0$ these two equations read $\begin{pmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \mathbf{0}$ with a matrix of determinant $W(t_0) \neq 0$, which is invertible; multiplying by its inverse gives $\alpha = \beta = 0$.

(2) "If": part (1). "Only if": the proof of the previous theorem showed that if $W(t_0) = 0$ at any single point (equation (11.16)) the solutions are proportional. So for independent solutions W never vanishes; for dependent ones it vanishes identically.

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Abel's identity (Robinson Exercise 11.2)

Differentiate $W = x_1\dot{x}_2 - x_2\dot{x}_1$ with the product rule:

$$\dot{W} = \dot{x}_1\dot{x}_2 + x_1\ddot{x}_2 - \dot{x}_2\dot{x}_1 - x_2\ddot{x}_1 = x_1\ddot{x}_2 - x_2\ddot{x}_1.$$

Both are solutions, so $\ddot{x}_j = -p\dot{x}_j - qx_j$. Substituting,

$$\dot{W} = x_1\big(-p\dot{x}_2 - qx_2\big) - x_2\big(-p\dot{x}_1 - qx_1\big) = -p\left(x_1\dot{x}_2 - x_2\dot{x}_1\right) - q\left(x_1x_2 - x_2x_1\right) = -p(t)\,W.$$

This first-order linear equation (Topic 2) has the solution

ABEL'S IDENTITY

$$W(t) = W(t_0)\,\exp\!\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

The exponential is never zero, so W is either identically zero or never zero on I. With constant coefficients p, $W(t) = W(t_0)e^{-p(t-t_0)}$.

THE CONVERSE OF (1) FAILS FOR ARBITRARY FUNCTIONS (ROBINSON EXERCISE 11.3)
$f_1 = t^2|t|$ and $f_2 = t^3$ are linearly independent on $\mathbb{R}$ (they agree for $t > 0$ and are opposite for $t < 0$, so no single constant relates them), yet $\dot{f}_1 = 3t|t|$ and $W = t^2|t| \cdot 3t^2 - t^3 \cdot 3t|t| = 0$ for every t. A zero Wronskian proves dependence only for solutions of a linear ODE.

SECTION 03 • ROBINSON CH. 12, SEC. 16.1 • LECTURE SEP 22-24

3. Constant coefficients: the three root cases

For the homogeneous equation with constant coefficients [Eq. (12.1), p. 111]

$$a\ddot{x} + b\dot{x} + cx = 0, \qquad a \neq 0, \tag{12.1}$$

try $x = e^{kt}$: it is the solution of $\dot{x} = kx$, and differentiating it only multiplies by k. With $\frac{d}{dt}e^{kt} = ke^{kt}$ and $\frac{d^2}{dt^2}e^{kt} = k^2e^{kt}$ (chain rule), substitution gives $(ak^2 + bk + c)e^{kt} = 0$. Since $e^{kt} \neq 0$:

$$ak^2 + bk + c = 0 \qquad \text{(auxiliary or characteristic equation)}, \qquad k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{12.2}$$

3.1 Case 1: two distinct real roots, $b^2 > 4ac$ [Sec. 12.1, p. 112]

Both $e^{k_1 t}$ and $e^{k_2 t}$ are solutions, and their Wronskian is $W = e^{k_1 t}\, k_2 e^{k_2 t} - e^{k_2 t}\, k_1 e^{k_1 t} = (k_2 - k_1)e^{(k_1+k_2)t} \neq 0$. Hence

$$x(t) = Ae^{k_1 t} + Be^{k_2 t}.$$

EXAMPLE 12.1 (P. 112): $\ddot{x} + \dot{x} - 6x = 0$, $x(0) = 1$, $\dot{x}(0) = 2$
$k^2 + k - 6 = (k-2)(k+3) = 0$, so $k = 2, -3$ and $x = Ae^{2t} + Be^{-3t}$, $\dot{x} = 2Ae^{2t} - 3Be^{-3t}$. At $t = 0$: $A + B = 1$, $2A - 3B = 2$. Subtracting twice the first, $-5B = 0$, so $B = 0$, $A = 1$: $x(t) = e^{2t}$.

If one root is 0 (e.g. $\ddot{x} - 2\dot{x} = 0$, $k = 0, 2$), the corresponding solution is the constant $e^{0t} = 1$: $x = A + Be^{2t}$ (Example 12.2).

3.2 Case 2: a repeated root, $b^2 = 4ac$ [Sec. 12.2, pp. 113–115]

With $\lambda = -b/(2a)$ and $c/a = b^2/(4a^2) = \lambda^2$, the equation is $\ddot{x} - 2\lambda\dot{x} + \lambda^2 x = 0$ (12.5), and $k^2 - 2\lambda k + \lambda^2 = (k-\lambda)^2 = 0$ gives only $e^{\lambda t}$. The second solution is $te^{\lambda t}$. Verification (product rule):

$$\dot{x} = e^{\lambda t} + \lambda t e^{\lambda t}, \qquad \ddot{x} = \lambda e^{\lambda t} + \lambda e^{\lambda t} + \lambda^2 t e^{\lambda t} = 2\lambda e^{\lambda t} + \lambda^2 t e^{\lambda t},$$

$$\ddot{x} - 2\lambda\dot{x} + \lambda^2 x = \big(2\lambda + \lambda^2 t - 2\lambda - 2\lambda^2 t + \lambda^2 t\big)e^{\lambda t} = 0 \ \checkmark.$$

Their Wronskian is $e^{\lambda t}(e^{\lambda t} + \lambda t e^{\lambda t}) - t e^{\lambda t}\lambda e^{\lambda t} = e^{2\lambda t} \neq 0$, so (12.6): $x(t) = (A + Bt)e^{\lambda t}$. Section 7 *derives* $te^{\lambda t}$ by reduction of order.

EXAMPLE 12.3 (PP. 114–115): $\ddot{x} + 2\dot{x} + x = 0,\ x(0) = 0,\ \dot{x}(0) = 1$

$(k+1)^2 = 0, k = -1$ twice. $x = (A + Bt)e^{-t}$, $\dot{x} = (B - A - Bt)e^{-t}$. Then $A = 0$ and $B - A = 1$: $x(t) = te^{-t}$.

3.3 Case 3: complex conjugate roots, $b^2 < 4ac$ [Sec. 12.3, pp. 115–117]

The roots are $k = \rho \pm i\omega$ with $\rho = -b/(2a)$, $\omega = \sqrt{4ac - b^2}/(2a)$. The complex solutions combine into real ones:

1. For real x take $x = Ce^{(\rho+i\omega)t} + C^* e^{(\rho - i\omega)t}$ (12.8), $C = \alpha + i\beta$; the two terms are complex conjugates, so $x = 2\,\mathrm{Re}\big[Ce^{(\rho+i\omega)t}\big]$.
2. Euler's formula $e^{i\omega t} = \cos\omega t + i\sin\omega t$: $Ce^{(\rho+i\omega)t} = e^{\rho t}(\alpha + i\beta)(\cos\omega t + i\sin\omega t)$, whose real part is $e^{\rho t}(\alpha\cos\omega t - \beta\sin\omega t)$.
3. With $A = 2\alpha$, $B = -2\beta$ (arbitrary reals):

$$x(t) = e^{\rho t}\big(A\cos\omega t + B\sin\omega t\big). \tag{12.9}$$

$\rho < 0$: decaying oscillation; $\rho = 0$: pure oscillation (simple harmonic motion); $\rho > 0$: growing oscillation. The Wronskian of $e^{\rho t}\cos\omega t$ and $e^{\rho t}\sin\omega t$ is $\omega e^{2\rho t} \neq 0$ (Exercise 11.1 (iii)).

EXAMPLE 12.4 (PP. 117–118): $\ddot{x} + 2\dot{x} + 5x = 0,\ x(0) = 1,\ \dot{x}(0) = 0$

$k = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$, so $x = e^{-t}(A\cos 2t + B\sin 2t)$ and, by the product rule, $\dot{x} = e^{-t}\big[(2B - A)\cos 2t - (2A + B)\sin 2t\big]$. Then $A = 1, 2B - A = 0$, so $B = \frac{1}{2}$: $x(t) = e^{-t}\big(\cos 2t + \frac{1}{2}\sin 2t\big)$.

ROOTS OF $ak^2 + bk + c = 0$	FUNDAMENTAL PAIR	GENERAL SOLUTION
Real, $k_1 \neq k_2$	$e^{k_1 t}, e^{k_2 t}$	$Ae^{k_1 t} + Be^{k_2 t}$
Repeated, k	e^{kt}, te^{kt}	$(A + Bt)e^{kt}$
Complex, $\rho \pm i\omega$	$e^{\rho t}\cos\omega t,\ e^{\rho t}\sin\omega t$	$e^{\rho t}(A\cos\omega t + B\sin\omega t)$

HIGHER ORDER: THE SAME STRATEGY (ROBINSON SEC. 16.1, LECTURE SEP 24)

For $a_n x^{(n)} + \cdots + a_1 \dot{x} + a_0 x = 0$ the trial e^{kt} gives $a_n k^n + \cdots + a_0 = 0$. Each simple real root contributes e^{kt}; a root repeated m times contributes $e^{kt}, te^{kt}, \ldots, t^{m-1}e^{kt}$; each complex pair $\rho \pm i\omega$ contributes $e^{\rho t}\cos\omega t$, $e^{\rho t}\sin\omega t$ (times $1, t, \ldots$ if repeated). The homogeneous part of Example 16.2 ($x^{(4)} - 16x = 64\sin 2t$): $x^{(4)} - 16x = 0$ has $k^4 = 16$, $k = \pm 2, \pm 2i$, so $x = Ae^{2t} + Be^{-2t} + C\cos 2t + D\sin 2t$.

SECTION 04 • ROBINSON CH. 13 • LECTURE SEP 22

4. The mass–spring–damper

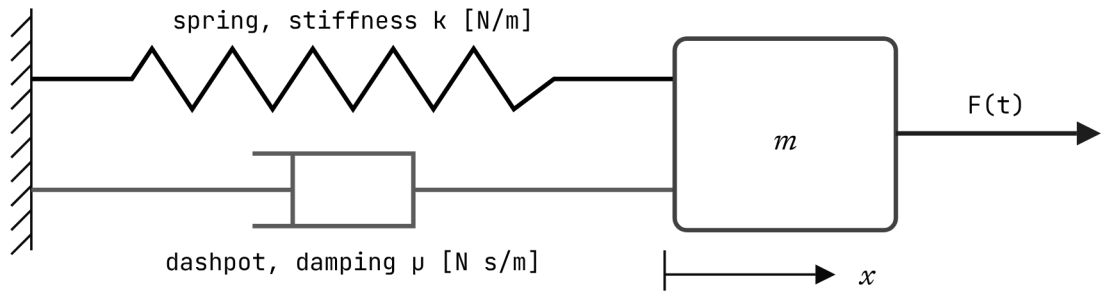


Figure 3.1. *Mass–spring–damper. The displacement x is measured from the equilibrium position, positive to the right. The spring pushes back with $-kx$ (Hooke's law), the dashpot with $-\mu\dot{x}$, and an external force $F(t)$ may act on the mass.*

4.1 Model

Hypotheses. A rigid mass m moves along one axis (one degree of freedom x, measured from equilibrium). The spring obeys Hooke's law, restoring force $-kx$; the dashpot gives a force proportional and opposite to the velocity, $-\mu\dot{x}$, with $\mu > 0$ [Robinson Sec. 13.1, p. 120; Sec. 13.3, p. 123]. Newton's second law:

$$m\ddot{x} = -kx - \mu\dot{x} + F(t) \quad \Longleftrightarrow \quad m\ddot{x} + \mu\dot{x} + kx = F(t).$$

SYMBOL	MEANING	SI UNIT
m	Mass	kg
k	Spring stiffness	N/m = kg/s²
μ	Damping coefficient	N s/m = kg/s
$\omega = \sqrt{k/m}$	Natural angular frequency	rad/s
$\lambda = \mu/m$	Damping rate	s^{-1}

Dividing by m, the free ($F = 0$) equation is [Eq. (13.6), p. 124]

$$\ddot{x} + \lambda\dot{x} + \omega^2 x = 0, \qquad \lambda = \frac{\mu}{m} > 0, \quad \omega^2 = \frac{k}{m}. \tag{13.6}$$

> **ERRATUM — ROBINSON P. 124**
> The text says "dividing by m ... and setting $\lambda = \mu/k$". Dividing $m\ddot{x} + \mu\dot{x} + kx = 0$ by m gives the coefficient μ/m for $\dot{x}$, so the correct definition is $\lambda = \mu/m$. Dimensional check: $[\mu/m] = (\mathrm{kg/s})/\mathrm{kg} = \mathrm{s}^{-1}$, as required for $\lambda\dot{x}$ to have the units of $\ddot{x}$; $[\mu/k] = (\mathrm{kg/s})/(\mathrm{kg/s}^2) = \mathrm{s}$ would not.

4.2 No damping: simple harmonic motion

With $\lambda = 0$: $\ddot{x} = -\omega^2 x$ (13.1), roots $\pm i\omega$, so $x = A\cos\omega t + B\sin\omega t$ (13.2). Writing $M = \sqrt{A^2 + B^2}$ and choosing ϕ with $\cos\phi = A/M$, $\sin\phi = B/M$:

$$x = M\big(\cos\phi\cos\omega t + \sin\phi\sin\omega t\big) = M\cos(\omega t - \phi). \tag{13.3}$$

The motion has amplitude M, period $2\pi/\omega = 2\pi\sqrt{m/k}$ independent of the amplitude, and lags $M\cos\omega t$ by the time ϕ/ω. The same equation governs the small oscillations of a pendulum, $\ddot{\theta} = -(g/L)\theta$, with period $2\pi\sqrt{L/g}$ (Sec. 13.2).

4.3 The three damping regimes [Sec. 13.3, pp. 124–126]

The auxiliary equation $k^2 + \lambda k + \omega^2 = 0$ has roots $k = \tfrac{1}{2}\big(-\lambda \pm \sqrt{\lambda^2 - 4\omega^2}\big)$.

- **Over-damped, $\lambda > 2\omega$:** two real roots $-k_1, -k_2$, both negative because $0 < \sqrt{\lambda^2 - 4\omega^2} < \lambda$. $x = Ae^{-k_1 t} + Be^{-k_2 t}$: decay with no oscillation.
- **Critically damped, $\lambda = 2\omega$:** repeated root $-\lambda/2 = -\omega$. $x = (A + Bt)e^{-\omega t}$: no oscillation, and the fastest return to equilibrium; any less damping allows oscillation.
- **Under-damped, $\lambda < 2\omega$:** roots $-\lambda/2 \pm i\sigma$ with $\sigma = \tfrac{1}{2}\sqrt{4\omega^2 - \lambda^2}$.

$$x = e^{-\lambda t/2}\big(A\cos\sigma t + B\sin\sigma t\big) = Me^{-\lambda t/2}\cos(\sigma t - \phi). \tag{13.7}$$

The amplitude decays exponentially and successive maxima are separated by the constant time $2\pi/\sigma$: setting $\dot{x} = -Me^{-\lambda t/2}\big[\tfrac{\lambda}{2}\cos(\sigma t - \phi) + \sigma\sin(\sigma t - \phi)\big] = 0$ gives $\tan(\sigma t - \phi) = -\lambda/(2\sigma)$, whose solutions are spaced by π/σ (alternating maxima and minima).

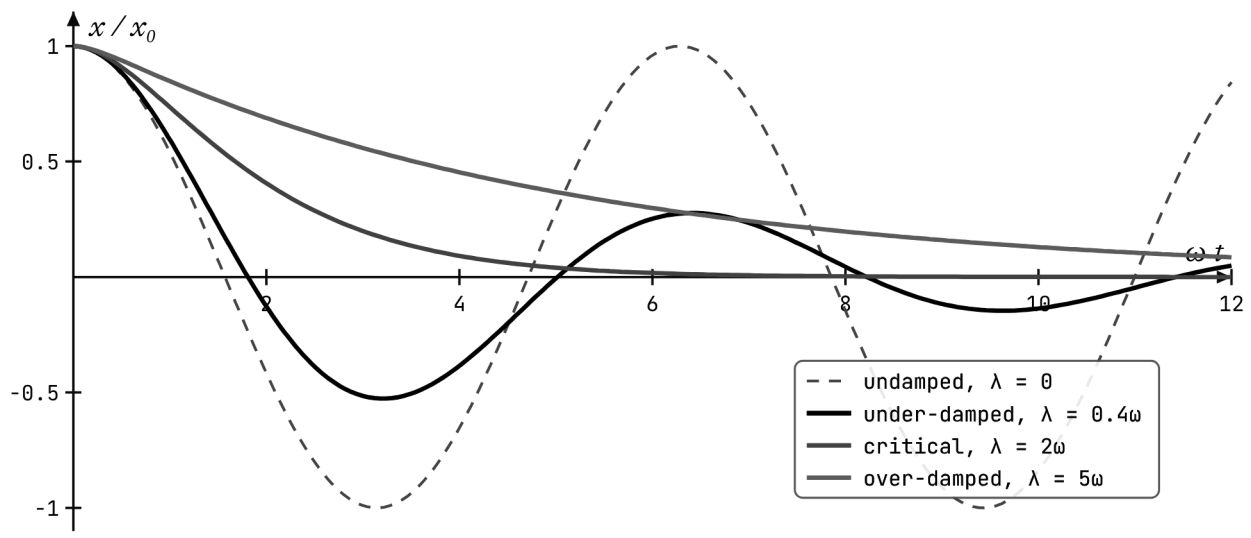


Figure 3.2. *Free response of $\ddot{x} + \lambda\dot{x} + \omega^2 x = 0$ released from rest at x_0: undamped (dashed), under-damped (indigo, decaying oscillation), critically damped (green, fastest return without overshoot) and over-damped (amber, slow creep). Compare Robinson Fig. 13.5.*

ENGINEERING NOTATION (COMPLEMENTARY)

Engineers write $\lambda = 2\zeta\omega$, i.e. $\zeta = \lambda/(2\omega) = \mu/(2\sqrt{km})$, the **damping ratio**: $\zeta < 1$ under-damped, $\zeta = 1$ critical, $\zeta > 1$ over-damped. The critical damping coefficient is $\mu_c = 2\sqrt{km}$ and the damped frequency is $\sigma = \omega\sqrt{1-\zeta^2}$.

A LIGHTLY DAMPED STRUCTURE (ROBINSON EXERCISE 13.8, THE MILLENNIUM BRIDGE)

Model $M\ddot{x} + k\dot{x} + \lambda x = 0$ with $M \approx 4 \times 10^5$ kg, damping $k \approx 5 \times 10^4$ kg/s, stiffness $\lambda \approx 10^7$ kg/s^2 (the exercise uses k and λ the other way round from Sec. 13.3). Natural frequency $\sqrt{\lambda/M} = \sqrt{25} = 5$ rad/s, i.e. $5/(2\pi) \approx 0.8$ Hz. Critical damping would be $2\sqrt{M\lambda} = 2\sqrt{4\times10^{12}} = 4 \times 10^6$ kg/s, so the actual damping is $5 \times 10^4 / 4 \times 10^6 \approx 1.3\%$ of critical. Pedestrians walking in step added a force $\approx 300\dot{x}$ each, i.e. *negative* damping: with enough people the net coefficient of $\dot{x}$ changes sign and oscillations grow.

AEROSPACE CONNECTION: LANDING GEAR (ILLUSTRATIVE NUMBERS)

A landing-gear leg carrying $m = 2000$ kg on an oleo of stiffness $k = 2 \times 10^5$ N/m has $\omega = \sqrt{k/m} = 10$ rad/s and $\mu_c = 2\sqrt{km} = 2\sqrt{4\times10^8} = 4 \times 10^4$ N s/m. Designers choose ζ of the order of 0.5–0.7 to avoid repeated bouncing (under-damping) without making the strut too stiff in compression (over-damping). With $\zeta = 0.6$: $\lambda = 12$ s^{-1}, $\sigma = 10\sqrt{1-0.36} = 8$ rad/s, and the amplitude falls by $e^{-\lambda T/2} = e^{-6\cdot 2\pi/8} \approx 0.009$ per damped period $T = 2\pi/\sigma$.

5. Inhomogeneous equations: undetermined coefficients

LEMMA — STRUCTURE OF THE GENERAL SOLUTION [ROBINSON, SEC. 14.1, PP. 131-132]

Let $y(t) = Ay_1(t) + By_2(t)$ be the general solution of $L[y] = 0$ (the **complementary function**) and x_p any one solution of $L[x_p] = f$ (a **particular integral**). Then the general solution of $L[x] = f$ is

$$x(t) = Ay_1(t) + By_2(t) + x_p(t).$$

PROOF

It is a solution: by linearity $L[y + x_p] = L[y] + L[x_p] = 0 + f = f$. **Every solution has this form:** if $L[x] = f$, then $L[x - x_p] = L[x] - L[x_p] = f - f = 0$, so $x - x_p$ solves the homogeneous equation and equals $Ay_1 + By_2$ for some A, B (Section 2). The two constants fit any initial conditions.

∎

For constant coefficients and right-hand sides built from polynomials, exponentials, sines and cosines, a particular integral has the same form as f: guess it with unknown ("undetermined") coefficients and substitute [Sec. 14.2-14.5].

FORCING $f(t)$	STANDARD GUESS x_p	IF A TERM OF THE GUESS SOLVES $L[y] = 0$
Polynomial of degree n	$c_n t^n + \cdots + c_1 t + c_0$ (all powers)	Multiply by t (e.g. root $k = 0$)
$ce^{\kappa t}$	$Ce^{\kappa t}$	$Cte^{\kappa t}$; $Ct^2e^{\kappa t}$ if κ is a repeated root
$\alpha \sin \sigma t + \beta \cos \sigma t$	$C \sin \sigma t + D \cos \sigma t$ (both, always)	$t(C \sin \sigma t + D \cos \sigma t)$ — resonance
Sum $\alpha f_1 + \beta f_2$	$\alpha x_1 + \beta x_2$ with $L[x_j] = f_j$	Treat each part separately (14.7)
Products, e.g. $te^{-t} \cos 2t$	Product of the guesses: $(At + B)e^{-t} \cos 2t + (Ct + D)e^{-t} \sin 2t$	As above

RULE OF THUMB [ROBINSON, SEC. 14.5, P. 139]

Take the standard guess. If it contains terms that satisfy the homogeneous equation, multiply the guess by t; repeat until no term of the guess solves the homogeneous equation.

EXAMPLE 14.4 (P. 134): POLYNOMIAL FORCING, $\ddot{x} + \dot{x} - 6x = 216t^3$

The complementary function is $Ae^{2t} + Be^{-3t}$. Try the full cubic $x_p = Ct^3 + Dt^2 + Et + F$, so $\dot{x}_p = 3Ct^2 + 2Dt + E$ and $\ddot{x}_p = 6Ct + 2D$. Substituting and collecting powers:

$$-6Ct^3 + (3C - 6D)t^2 + (6C + 2D - 6E)t + (2D + E - 6F) = 216t^3.$$

Matching coefficients from the top: $-6C = 216 \Rightarrow C = -36$; $3C - 6D = 0 \Rightarrow D = -18$; $6C + 2D - 6E = 0 \Rightarrow E = (-216 - 36)/6 = -42$; $2D + E - 6F = 0 \Rightarrow F = (-36 - 42)/6 = -13$. Hence $x = Ae^{2t} + Be^{-3t} - 36t^3 - 18t^2 - 42t - 13$. A bare Ct^3 would fail: its derivatives generate t^2, t and constant terms that must be cancelled.

EXAMPLE 14.8 (P. 136): EXPONENTIAL FORCING THAT HITS THE COMPLEMENTARY FUNCTION, $\ddot{x} + \dot{x} - 6x = 5e^{-3t}$

e^{-3t} already solves the homogeneous equation, so Ce^{-3t} gives $0 = 5e^{-3t}$. Multiply by t: $x_p = Cte^{-3t}$, with $\dot{x}_p = C(1 - 3t)e^{-3t}$ and $\ddot{x}_p = C(-6 + 9t)e^{-3t}$ (product rule twice). Then

$$\ddot{x}_p + \dot{x}_p - 6x_p = C\big[(-6 + 9t) + (1 - 3t) - 6t\big]e^{-3t} = -5Ce^{-3t} = 5e^{-3t} \;\Rightarrow\; C = -1.$$

The te^{-3t} terms cancel, as they must. $x = Ae^{2t} + Be^{-3t} - te^{-3t}$.

EXAMPLE 14.10 (PP. 137–138): SINUSOIDAL FORCING, $\ddot{x} + 2\dot{x} + x = 100\cos 2t$

Complementary function $(A + Bt)e^{-t}$. Try $x_p = C\sin 2t + D\cos 2t$ (both terms: the $\dot{x}$ term turns cosines into sines). $\dot{x}_p = 2C\cos 2t - 2D\sin 2t$, $\ddot{x}_p = -4C\sin 2t - 4D\cos 2t$, so

$$\ddot{x}_p + 2\dot{x}_p + x_p = (4C - 3D)\cos 2t - (3C + 4D)\sin 2t = 100\cos 2t.$$

$3C + 4D = 0$ and $4C - 3D = 100$ give $C = 16$, $D = -12$ (check: $64 + 36 = 100$, $48 - 48 = 0$). So $x = (A + Bt)e^{-t} + 16\sin 2t - 12\cos 2t$. As $t \to \infty$ the transient dies and the steady oscillation has amplitude $\sqrt{16^2 + 12^2} = 20$.

EXAMPLE 14.11 (P. 138): RESONANCE, $\ddot{x} + x = 8\cos t$

$\cos t$ and $\sin t$ solve the homogeneous equation, so try $x_p = t(C\sin t + D\cos t)$. Differentiating twice with the product rule, $\ddot{x}_p = 2C\cos t - 2D\sin t - t(C\sin t + D\cos t)$, so $\ddot{x}_p + x_p = 2C\cos t - 2D\sin t = 8\cos t$: $C = 4$, $D = 0$. $x = A\sin t + B\cos t + 4t\sin t$: the forced part grows linearly in time.

SECTION 06 • ROBINSON CH. 15

6. Forced vibrations and resonance

Force the oscillator at angular frequency α: $\ddot{x} + \omega^2 x = a\cos\alpha t$ (15.3).

- $\alpha \neq \omega$ [Sec. 15.1.1]: try $x_p = C\cos\alpha t$ (no sine needed – there is no $\dot{x}$ term): $(-\alpha^2+\omega^2)C = a$, so $x = M\cos(\omega t - \phi) + \dfrac{a}{\omega^2-\alpha^2}\cos\alpha t$ (15.4). Bounded, but large when $\alpha \approx \omega$. From rest, $x = \dfrac{\cos\alpha t - \cos\omega t}{\omega^2-\alpha^2}\cdot a = \dfrac{2a}{\omega^2-\alpha^2}\sin\dfrac{(\omega+\alpha)t}{2}\sin\dfrac{(\omega-\alpha)t}{2}$: a fast oscillation modulated by a slow envelope – **beats** (Exercises 15.1–15.2).
- $\alpha = \omega$ [Sec. 15.1.2]: the rule of thumb gives $x_p = \dfrac{a}{2\omega}\,t\sin\omega t$: the amplitude grows without bound. This is **resonance** (Robinson recalls the 1831 Broughton bridge collapse under marching troops). The same result follows from the beats formula by letting $\alpha \to \omega$ with L'Hôpital's rule (Ex. 15.3).

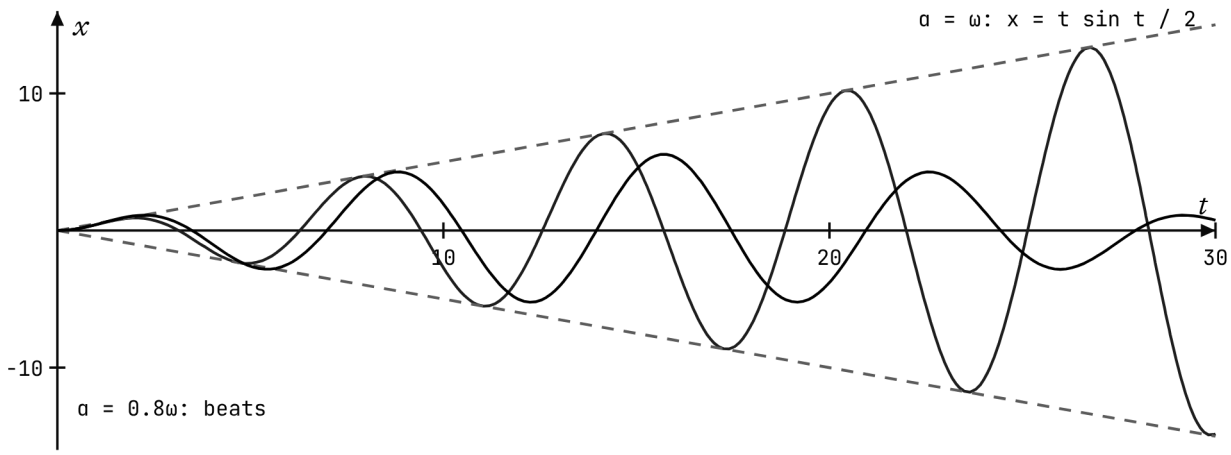


Figure 3.3. *Undamped forcing $\ddot{x}+\omega^2 x = \cos\alpha t$, $\omega = 1$, from rest. Red: exact resonance $\alpha = \omega$, amplitude growing linearly, $x = t\sin t/2$ (Robinson Sec. 15.1.2, Ex. 15.3). Indigo: $\alpha = 0.8$, bounded beats $x = (\cos\alpha t - \cos\omega t)/(\omega^2-\alpha^2)$ (Ex. 15.1–15.2).*

6.1 Damped forcing: the frequency response [Sec. 15.2, pp. 145–148]

For $\ddot{x} + \lambda\dot{x} + \omega^2 x = a\cos\alpha t$ with $\lambda > 0$, try $x_p = C\sin\alpha t + D\cos\alpha t$. Matching sines and cosines:

$$C(\omega^2-\alpha^2) = D\lambda\alpha, \qquad D(\omega^2-\alpha^2) + C\lambda\alpha = a.$$

Solving (substitute C from the first into the second):

$$C = \frac{a\lambda\alpha}{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}, \qquad D = \frac{a(\omega^2-\alpha^2)}{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}, \qquad x_p = C\sin\alpha t + D\cos\alpha t. \tag{15.5}$$

The complementary function decays (all roots have negative real part), so after a transient only x_p remains, with amplitude

$$R(\alpha) = \sqrt{C^2+D^2} = \frac{a}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}} = a\,F(\alpha), \qquad F(\alpha) = \frac{1}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}}. \tag{15.6}$$

The denominator is smallest where $\frac{d}{d\alpha}\left[(\omega^2-\alpha^2)^2+\lambda^2\alpha^2\right] = -4\alpha(\omega^2-\alpha^2)+2\lambda^2\alpha = 0$ (chain rule on the square), i.e. $2\alpha\,(2\alpha^2 - 2\omega^2 + \lambda^2) = 0$. For $\lambda^2 < 2\omega^2$ this gives an interior peak at $\alpha^2 = \omega^2 - \tfrac{1}{2}\lambda^2$, where $R_{\max} = \dfrac{2a}{\lambda\sqrt{4\omega^2-\lambda^2}} \approx \dfrac{a}{\lambda\omega}$ for small λ. The response is finite for any $\lambda > 0$ but inversely proportional to the damping: **pseudo-resonance**. For $\lambda^2 \geq 2\omega^2$ there is no peak: $R(\alpha)$ decreases monotonically from $R(0) = a/\omega^2$.

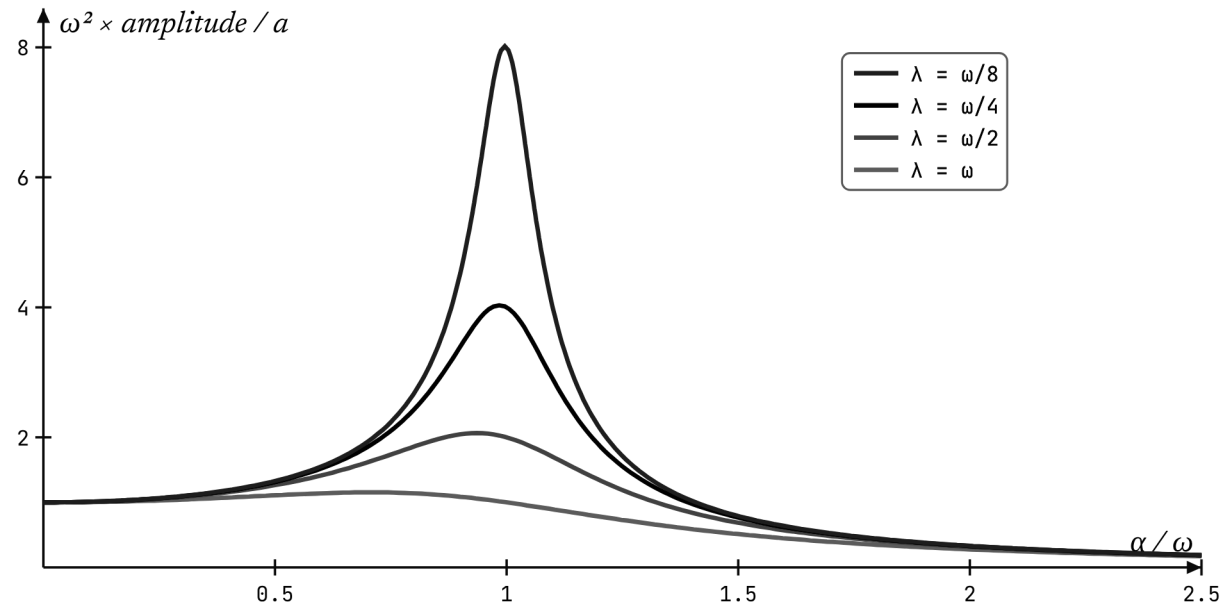


Figure 3.4. *Steady-state amplitude of $\ddot{x} + \lambda\dot{x} + \omega^2 x = a\cos\alpha t$, scaled by a/ω^2: $\omega^2 F(\alpha) = \omega^2 / \sqrt{(\omega^2-\alpha^2)^2 + \lambda^2\alpha^2}$ (Robinson (15.6), Fig. 15.5). The peak sharpens and moves towards $\alpha = \omega$ as damping decreases.*

- **Limits:** $\alpha \to 0$: $R \to a/\omega^2$ (static deflection under a steady force); $\alpha \to \infty$: $R \approx a/\alpha^2 \to 0$ (inertia dominates).
- **Dimensions:** a is force per unit mass [m/s²]; $[\omega^2 - \alpha^2] = [\lambda\alpha] = \mathrm{s}^{-2}$, so $[R] = \mathrm{m}$.

AEROSPACE CONNECTION

Rotor blades, engine mounts and wing structures are tuned so that their natural frequencies stay away from forcing frequencies such as rotor or shaft speeds. Helicopter *ground resonance* and aeroelastic *flutter* are more dangerous still: as in the Millennium Bridge example, the coupling with the excitation acts as negative damping, and the amplitude grows exponentially rather than linearly.

SECTION 07 • ROBINSON CH. 17 • LECTURE SEP 24–29

7. Variable coefficients: reduction of order

For $a(t)\ddot{x} + b(t)\dot{x} + c(t)x = 0$ there is no general recipe for solutions. But if **one** solution $u(t)$ is known, a second one follows from a first-order equation [Robinson, Ch. 17, pp. 159–160].

1. Substitute $x = u(t)\,y(t)$ (the constant A in Au is promoted to a function). Product rule: $\dot{x} = \dot{u}y + u\dot{y},\ \ddot{x} = \ddot{u}y + 2\dot{u}\dot{y} + u\ddot{y}$.

2. Substitute and group by $y, \dot{y}, \ddot{y}$:

$$y\left[a\ddot{u} + b\dot{u} + cu\right] + a\big(2\dot{u}\dot{y} + u\ddot{y}\big) + bu\dot{y} = 0.$$

   The bracket is zero because u is a solution – this is why the method works.

3. y itself has disappeared. With $z = \dot{y}$ we get the first-order linear equation $\big[au\big]\dot{z} + \big[2a\dot{u} + bu\big]z = 0$ (17.3).

4. Solve for z (separable or integrating factor), integrate once to get y, and set $x = uy$.

In the normalised form $\ddot{x} + p\dot{x} + qx = 0$ the result is $z = \dfrac{A\,e^{-\int p\,dt}}{u^2}$ (Robinson Exercise 17.7), so $x_2 = u\displaystyle\int \frac{e^{-\int p\,dt}}{u^2}\,dt$. Robinson advises applying the method rather than memorising this formula.

EXAMPLE 17.1 (PP. 160–161): DERIVING THE REPEATED-ROOT SOLUTION

For $\ddot{x} - 2\lambda\dot{x} + \lambda^2 x = 0$ with known $u = e^{\lambda t}$, set $x = ye^{\lambda t}$: $\dot{x} = (\lambda y + \dot{y})e^{\lambda t}$, $\ddot{x} = (\lambda^2 y + 2\lambda\dot{y} + \ddot{y})e^{\lambda t}$. Substituting and cancelling $e^{\lambda t} \neq 0$:

$$\lambda^2 y + 2\lambda\dot{y} + \ddot{y} - 2\lambda^2 y - 2\lambda\dot{y} + \lambda^2 y = \ddot{y} = 0 \implies y = A + Bt \implies x = (A + Bt)e^{\lambda t}. \quad (17.4)$$

The second solution $te^{\lambda t}$ of Section 3.2 is now derived, not guessed.

EXAMPLE 17.2 (PP. 161–162): $t^2\ddot{x} - 2t\dot{x} - 4x = 0$, $t > 0$, WITH $u = 1/t$

1. Check u: $\dot{u} = -t^{-2}$, $\ddot{u} = 2t^{-3}$, so $t^2(2t^{-3}) - 2t(-t^{-2}) - 4t^{-1} = (2 + 2 - 4)t^{-1} = 0$.
2. $x = y/t$: $\dot{x} = \dfrac{\dot{y}}{t} - \dfrac{y}{t^2}$, $\ddot{x} = \dfrac{\ddot{y}}{t} - \dfrac{2\dot{y}}{t^2} + \dfrac{2y}{t^3}$.
3. Substitute: $t\ddot{y} - 2\dot{y} + \dfrac{2y}{t} - 2\dot{y} + \dfrac{2y}{t} - \dfrac{4y}{t} = t\ddot{y} - 4\dot{y} = 0$.
4. $z = \dot{y}$: $\dot{z} - \dfrac{4}{t}z = 0$. Integrating factor $e^{-4\ln t} = t^{-4}$: $\frac{d}{dt}(t^{-4}z) = 0$, so $z = ct^4$.
5. $y = \displaystyle\int ct^4\,dt = \frac{c}{5}t^5 + b$ and $x = y/t = at^4 + b/t$. The new independent solution is $x_2 = t^4$.

Check: $t^2(12t^2) - 2t(4t^3) - 4t^4 = (12 - 8 - 4)t^4 = 0$. Wronskian: $t^{-1}\cdot 4t^3 - t^4\cdot(-t^{-2}) = 5t^2 \neq 0$ for $t > 0$.

8. Variation of parameters (Lagrange)

Undetermined coefficients needs constant coefficients and special right-hand sides. Variation of parameters works for any continuous forcing and any coefficients, provided two independent homogeneous solutions x_1, x_2 are known [Robinson, Ch. 18, pp. 164–166]. Consider

$$\ddot{x} + p(t)\dot{x} + q(t)x = g(t). \tag{18.4}$$

8.1 Derivation

1. **Ansatz.** Replace the constants of the complementary function by functions (18.5): $x(t) = u_1(t)x_1(t) + u_2(t)x_2(t)$.

2. **First derivative** (product rule, 18.6): $\dot{x} = \dot{u}_1 x_1 + u_1 \dot{x}_1 + \dot{u}_2 x_2 + u_2 \dot{x}_2$.

3. **Auxiliary condition.** Two unknown functions but one equation leaves one free condition. Choose it to stop second derivatives of u_j appearing:

$$\dot{u}_1 x_1 + \dot{u}_2 x_2 = 0, \tag{18.7}$$

   so that $\dot{x} = u_1 \dot{x}_1 + u_2 \dot{x}_2$.

4. **Second derivative:** $\ddot{x} = \dot{u}_1 \dot{x}_1 + u_1 \ddot{x}_1 + \dot{u}_2 \dot{x}_2 + u_2 \ddot{x}_2$.

5. **Substitute into (18.4)** and group:

$$u_1\left[\ddot{x}_1 + p\dot{x}_1 + qx_1\right] + u_2\left[\ddot{x}_2 + p\dot{x}_2 + qx_2\right] + \dot{u}_1\dot{x}_1 + \dot{u}_2\dot{x}_2 = g.$$

   Both brackets vanish (x_1, x_2 solve the homogeneous equation), leaving $\dot{u}_1\dot{x}_1 + \dot{u}_2\dot{x}_2 = g$.

6. **Linear system** for $\dot{u}_1$, $\dot{u}_2$:

$$\begin{pmatrix} x_1 & x_2 \\ \dot{x}_1 & \dot{x}_2 \end{pmatrix} \begin{pmatrix} \dot{u}_1 \\ \dot{u}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ g \end{pmatrix}.$$

   Its determinant is the Wronskian $W = x_1\dot{x}_2 - x_2\dot{x}_1$, never zero for independent solutions.

Cramer's rule:

$$\dot{u}_1 = \frac{\begin{vmatrix} 0 & x_2 \\ g & \dot{x}_2 \end{vmatrix}}{W} = -\frac{x_2\, g}{W}, \qquad \dot{u}_2 = \frac{\begin{vmatrix} x_1 & 0 \\ \dot{x}_1 & g \end{vmatrix}}{W} = \frac{x_1\, g}{W}.$$

7 **Integrate** and substitute into the ansatz:

VARIATION OF CONSTANTS FORMULA [ROBINSON, EQ. (18.8), P. 166]

$$x_p(t) = -x_1(t)\int \frac{x_2(t)\, g(t)}{W(t)}\, dt + x_2(t)\int \frac{x_1(t)\, g(t)}{W(t)}\, dt. \tag{18.8}$$

Constants of integration only add multiples of x_1, x_2, which belong to the complementary function.

EXAMPLE 18.1 (PP. 166–167): $\ddot{x}+\dot{x}-6x=5e^{-3t}$ AGAIN

$x_1 = e^{-3t}, x_2 = e^{2t}, W = e^{-3t}\cdot 2e^{2t} - e^{2t}\cdot(-3e^{-3t}) = 5e^{-t}$. Then $\dfrac{x_2 g}{W} = \dfrac{e^{2t}\cdot 5e^{-3t}}{5e^{-t}} = 1$ and $\dfrac{x_1 g}{W} = \dfrac{e^{-3t}\cdot 5e^{-3t}}{5e^{-t}} = e^{-5t}$, so

$$x_p = -e^{-3t}\int 1\, dt + e^{2t}\int e^{-5t}\, dt = -te^{-3t} + e^{2t}\left(-\frac{e^{-5t}}{5}\right) = -te^{-3t} - \frac{1}{5}e^{-3t}.$$

The second term is a multiple of x_1 and is absorbed into Ae^{-3t}: the particular integral $-te^{-3t}$ of Example 14.8 is recovered.

ERRATUM — ROBINSON P. 166

The book writes the second integral as $5e^{2t}\int e^{-5t}dt$ and the result as $-te^{-3t} - e^{-3t}$. The factor 5 was already cancelled by $W = 5e^{-t}$, so the term is $e^{2t}\cdot(-\frac{1}{5}e^{-5t}) = -\frac{1}{5}e^{-3t}$. Check: $\frac{d}{dt}\big(-\frac{1}{5}e^{-5t}\big) = e^{-5t}$, so $\int e^{-5t}dt = -\frac{1}{5}e^{-5t}$. The slip is harmless for the general solution, since any multiple of e^{-3t} is absorbed into the arbitrary constant, as Robinson notes.

EXAMPLE 18.2 (PP. 167–168): $\ddot{x}+x=\tan t$ — BEYOND UNDETERMINED COEFFICIENTS

1 $x_1 = \sin t, x_2 = \cos t$; ansatz $x = u\sin t + v\cos t$ with $\dot{u}\sin t + \dot{v}\cos t = 0$ (18.10).

2 Then $\dot{x} = u\cos t - v\sin t$, $\ddot{x} = \dot{u}\cos t - u\sin t - \dot{v}\sin t - v\cos t$, and the equation reduces to $\dot{u}\cos t - \dot{v}\sin t = \tan t$ (18.11).

3 Multiply (18.10) by $\sin t$, (18.11) by $\cos t$ and add: $\dot{u} = \tan t\cos t = \sin t$. Multiply (18.10) by $\cos t$, (18.11) by $\sin t$ and subtract: $\dot{v} = -\dfrac{\sin^2 t}{\cos t} = \cos t - \sec t$.

4. Integrate: $u = -\cos t$; $v = \sin t - \ln|\sec t + \tan t|$, using $\int \sec t\,dt = \ln|\sec t + \tan t|$ (differentiate to check: $\frac{\sec t \tan t + \sec^2 t}{\sec t + \tan t} = \sec t$).

5. $x_p = -\cos t \sin t + \big(\sin t - \ln|\sec t + \tan t|\big)\cos t = -\cos t\, \ln|\sec t + \tan t|$.

Check (write $\ell = \ln|\sec t + \tan t|$, $\dot\ell = \sec t$): $\dot x_p = \sin t\,\ell - 1$, $\ddot x_p = \cos t\,\ell + \sin t \sec t = \cos t\,\ell + \tan t$, so $\ddot x_p + x_p = \tan t$.

SECTION 09 • SUMMARY

9. Summary and lecture map

RESULT	STATEMENT	ROBINSON	LECTURE
Existence and uniqueness	Data $x(t_0)$, $\dot x(t_0)$ fix a unique solution	Thm 11.1, p. 102	Sep 22
Superposition	$\alpha x_1 + \beta x_2$ solves $L[x] = 0$	Sec. 11.2, p. 103	Sep 22
General solution	Two independent solutions, necessary and sufficient	Sec. 11.3, pp. 104–106	Sep 22
Wronskian	Independent $\iff W \neq 0$; Abel: $\dot W = -pW$	Sec. 11.4, Ex. 11.2	Sep 22
Constant coefficients	e^{kt}; three root cases	Ch. 12, Sec. 16.1	Sep 22–24
Mass–spring–damper	$\ddot x + \lambda\dot x + \omega^2 x = 0$; over/critical/under-damped	Ch. 13	Sep 22
Inhomogeneous	$x = y_h + x_p$; undetermined coefficients; rule of thumb	Ch. 14	Sep 24–29
Resonance	$x_p \propto t\sin\omega t$; damped peak at $\alpha^2 = \omega^2 - \lambda^2/2$ (if $\lambda^2 < 2\omega^2$)	Ch. 15	–
Reduction of order	$x = uy$ gives a first-order equation for $\dot y$	Ch. 17	Sep 24–29
Variation of parameters	Formula (18.8) with the Wronskian	Ch. 18	Oct 1

WHAT COMES NEXT

Topic 4 rewrites $\ddot x + p\dot x + qx = f$ as a first-order system for $(x, \dot x)$ and replaces "two independent solutions" by a fundamental matrix, the Wronskian by a determinant of vector solutions, and the auxiliary equation by an eigenvalue problem.